Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/31**

Paper 3 Practical Test **October/November 2023**

CONFIDENTIAL INSTRUCTIONS

3343095870

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials supplied by Cambridge International are confidential and must NOT reach candidates either directly or indirectly.

The supervisor must complete the report at the end of this document and return it with the scripts.

INSTRUCTIONS

- If you have any queries regarding these confidential instructions, contact Cambridge International stating the centre number, the syllabus and component number and the nature of the query.
  email info@cambridgeinternational.org
  phone +44 1223 553554

This document has **8** pages. Any blank pages are indicated.

DC (KN) 313488/3
© UCLES 2023

[Turn over

General information about practical exams

Centres must follow the guidance on science practical exams given in the *Cambridge Handbook*.

Safety

Supervisors must follow national and local regulations relating to safety and first aid.

Only those procedures described in the question paper should be attempted.

Supervisors must inform candidates that materials and apparatus used in the exam should be treated with caution. Suitable eye protection should be used where necessary.

The following hazard codes are used in these confidential instructions, where relevant:

C	corrosive	**MH**	moderate hazard
HH	health hazard	**T**	acutely toxic
F	flammable	**O**	oxidising
N	hazardous to the aquatic environment		

Hazard data sheets relating to substances used in this exam should be available from your chemical supplier.

Before the exam

- The packets containing the question papers must **not** be opened before the exam.
- It is assumed that standard school laboratory facilities, as indicated in the *Guide to Planning Practical Science*, will be available.
- Spare materials and apparatus for the tasks set must be available for candidates, if required.

During the exam

- It must be made clear to candidates at the start of the exam that they may request spare materials and apparatus for the tasks set.
- Where specified, the supervisor **must** perform the experiments and record the results as instructed. This must be done **out of sight** of the candidates, using the same materials and apparatus as the candidates.
- Any assistance provided to candidates must be recorded in the supervisor's report.
- If any materials or apparatus need to be replaced, for example, in the event of breakage or loss, this must be recorded in the supervisor's report.

After the exam

- The supervisor must complete a report for each practical session held and each laboratory used.
- Each packet of scripts returned to Cambridge International must contain the following items:
  - the scripts of the candidates specified on the bar code label provided
  - the supervisor's results relevant to these candidates
  - the supervisor's reports relevant to these candidates
  - seating plans for each practical session, referring to each candidate by candidate number
  - the attendance register.

© UCLES 2023

Specific information for this practical exam

During the exam, the supervisor (**not** the invigilator) must do the experiments in Questions 1 and 2 and record the results on a spare copy of the question paper, clearly labelled 'supervisor's results'.

Apparatus

The apparatus listed must be provided to each candidate.

Question 1

1 × plastic cup (expanded polystyrene / foamed plastic) which will fit into a 250 cm^3 beaker
1 × 250 cm^3 beaker
1 × 25 cm^3 measuring cylinder
1 × 10 cm^3 measuring cylinder
1 × 25 cm^3 pipette
1 × pipette filler
1 × 50 cm^3 burette
1 × stand
1 × burette clamp
1 × funnel for filling burette
1 × stirring thermometer, –10 °C to +110 °C with 1 °C graduations.
access to distilled water

Question 2

4 × test-tubes
1 × test-tube rack
1 × stopper to fit test-tubes
1 × boiling tube
1 × boiling tube holder suitable for heating
1 × wash bottle containing distilled water
1 × Bunsen burner and means to light it
wooden splints
a supply of teat/dropping pipettes
1 × beaker (for washing teat/dropping pipettes)
paper towels
red and blue litmus papers
apparatus normally used in the centre to test for carbon dioxide with limewater.

Candidates are expected to rinse and reuse test-tubes and boiling tubes where necessary.
Additional tubes should be available.

© UCLES 2023

© UCLES 2023

Materials

The materials listed in the table must be provided to each candidate. An excess of at least 10% of each material must be prepared to cover accidental loss.

Warning: small amounts of NH_3 **[C][T][N]**, which can cause respiratory distress in some people, may be produced. **The laboratory must be well ventilated.**

label		per candidate	identity	notes
Question 1				
[C]	**P**	200 cm^3	1.25 mol / dm^3 sodium hydroxide	Add 50.0 g of sodium hydroxide **[C]** to 500 cm^3 of distilled water. Dissolve the solid and then make the volume up to 1 dm^3 with distilled water.
[MH]	**Q**	150 cm^3	1.0 mol / dm^3 sulfuric acid	Cautiously pour 56 cm^3 of concentrated (98%) sulfuric acid **[C]** into 500 cm^3 of distilled water with continuous stirring. When cool, make the volume up to 1 dm^3 with distilled water.
	distilled water	100 cm^3	distilled water	
Supervisors are asked to carry out a standard acid/base titration between solutions **P** and **Q** to ensure that the concentrations of the two solutions fall within the required range. It is essential that 25.0 cm^3 of **P** reacts with between 15.0 cm^3 and 17.0 cm^3 of **Q**.				
Question 2				
	X	15 cm^3	0.2 mol / dm^3 aluminium ammonium sulfate solution, $NH_4Al(SO_4)_2$	Dissolve 4.74 g aluminium ammonium sulfate, $NH_4Al(SO_4)_2$, **[MH]** in 50 cm^3 distilled water and then make the volume up to 100 cm^3 with distilled water.
	Y	10 cm^3	1.0 mol / dm^3 hydrochloric acid	See preparation instructions in the 2023–25 syllabus.
	magnesium	2 × 1 cm	magnesium ribbon	

© UCLES 2023

label		per candidate	identity	notes
[C]	dilute nitric acid	10 cm^3	1.0 mol/dm^3 HNO_3	See preparation instructions in the 2023–25 syllabus. If necessary, each of these reagents can be provided as a communal supply for groups of up to 6 candidates. Invigilators must be alert to the risk of contamination and the opportunity for malpractice when using a communal supply.
[MH][N]	aqueous ammonia	15 cm^3	1.0 mol/dm^3 NH_3	
[C]	aqueous sodium hydroxide	20 cm^3	1.0 mol/dm^3 NaOH	
	aqueous silver nitrate	5 cm^3	0.05 mol/dm^3 $AgNO^3$	
[MH]	limewater	10 cm^3	saturated aqueous calcium hydroxide, $Ca(OH)_2$	

- All solutions must be thoroughly mixed.
- If you are unable to source any of these chemicals, you must contact Cambridge International as far as possible in advance of the exam for advice.
- Materials must be labelled only as specified in the 'label' column. The identities of chemicals labelled with letter codes, e.g. **P**, may be different from their descriptions in the question paper. Candidates must use the descriptions given in the question paper.
- If chemicals are prepared in more than one batch, clearly labelled supervisor results **must** be provided for each batch. The candidates using each batch must be listed on the supervisor's report.

[Turn over

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

© UCLES 2023

Supervisor's report

Syllabus and component number | | | | | / | | |

Centre number | | | | | |

Centre name .....

Time of the practical session .....

Laboratory name/number .....

Give details of any difficulties experienced by the centre or by candidates (include the relevant candidate names and candidate numbers).

You must include:

- any difficulties experienced by the centre in the preparation of materials
- any difficulties experienced by candidates, e.g. due to faulty materials or apparatus
- any specific assistance given to candidates.

© UCLES 2023

If chemicals have been prepared in more than one batch, list the candidates using each batch. Supervisor results must be prepared and submitted using each batch of chemicals.

Declaration

1 Each packet that I am returning to Cambridge International contains all of the following items:
- the scripts of the candidates specified on the bar code label provided
- the supervisor's results relevant to these candidates
- the supervisor's reports relevant to these candidates
- seating plans for each practical session, referring to each candidate by candidate number
- the attendance register.

2 Where the practical exam has taken place in more than one practical session, I have clearly labelled the supervisor's results, supervisor's reports and seating plans with the time and laboratory name/number for each practical session.

3 I have included details of difficulties relating to each practical session experienced by the centre or by candidates.

4 I have reported any other adverse circumstances affecting candidates, e.g. illness, bereavement or temporary injury, directly to Cambridge International on a *special consideration form*.

Signed ..... (supervisor)

Name (in block capitals) .....

© UCLES 2023